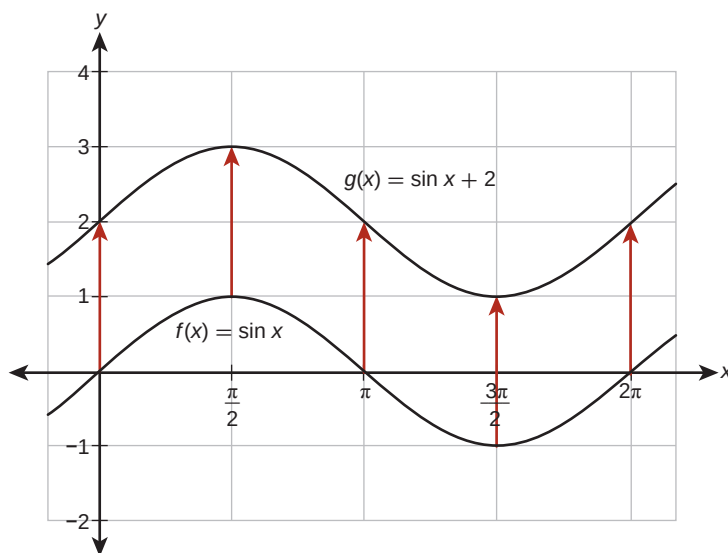


Sinusoidal Graphs: Vertical Shift

Adding a constant to a sinusoidal function causes a vertical shift.

Determining Vertical Shifts of the Sine Function

Compare the graph of $g(x) = \sin x + 2$ to $f(x) = \sin x$.



The graph of $g(x) = \sin x + 2$ is a translation up 2 units of $f(x) = \sin x$.

Vertical Translation of a Sine Function

Compared to the parent function $f(x) = \sin x$, $g(x) = \sin x + C$ is a vertical translation by C units such that

$C > 0$: shift up C units.

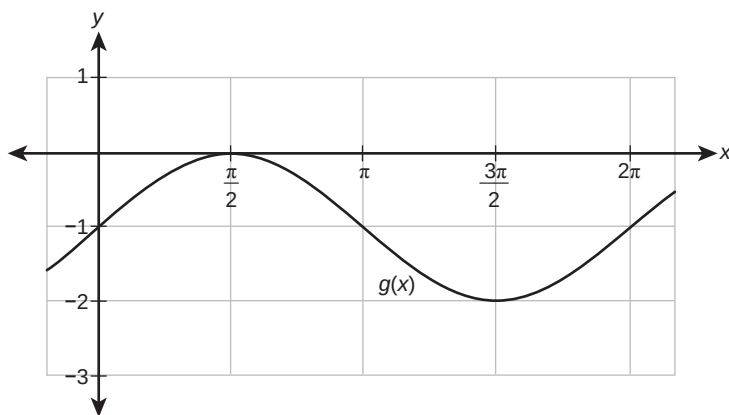
$C < 0$: shift down $|C|$ units.

In the graph of $g(x) = \sin x + 2$, the equation of the midline is $y = 2$.

For any function $g(x) = \sin x + C$, the equation of the midline is $y = C$.

EXAMPLE 1

Write an equation for the graph in the form $g(x) = \sin x + C$.



SOLUTION

Since the graph is a 1-unit shift down of $f(x) = \sin x$, $C = -1$. So the equation is $g(x) = \sin x - 1$. ■

Determining Vertical Shifts of the Cosine Function

As with a sine function, a vertical shift of a cosine function can be found using a formula.

Vertical Translation of a Cosine Function

Compared to the parent function $f(x) = \cos x$, $g(x) = \cos x + C$ is a vertical translation by C units, such that

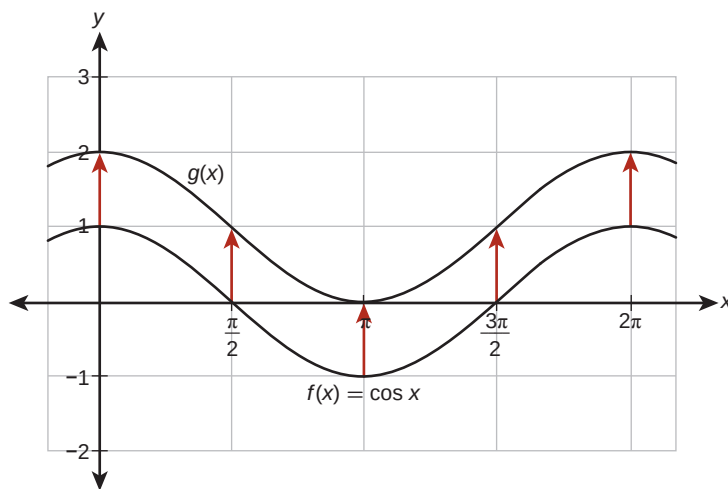
$C > 0$: shift up C units.

$C < 0$: shift down $|C|$ units.

Also, the equation of the midline of $g(x) = \cos x + C$ is $y = C$.

EXAMPLE 2

Use the graph of the cosine function as reference and the function $g(x)$ shown to write the equation of the function in the form $g(x) = \cos x + C$.



SOLUTION

Since the cosine function has been shifted up by 1 unit, $C = 1$. So the transformed function is $g(x) = \cos x + 1$. ■